

Donghyung Lee

DevOps Engineer | Resume & Career Description

E-mail: genius5711@gmail.com Blog(KR): somaz.tistory.com Blog(EN): somaz.blog GitHub: github.com/somaz94

Introduction

Hello, I am Donghyung Lee, a DevOps Engineer.

I have performed my work centered on 'standards and records' with the goal of improving infrastructure stability and efficiency. My greatest strength is the habit of thinking beyond simple operations, contemplating the meaning and purpose of each task, and seeking better directions.

I have experience in designing and building cloud infrastructure on AWS and GCP, and am proficient in service operations in Kubernetes (EKS, GKE, on-premises) environments, IaC implementation using Terraform and Ansible, and CI/CD pipeline construction with GitLab CI/CD, GitHub Actions, ArgoCD, and Jenkins. I also have experience building monitoring and logging systems based on PLG Stack and ELK Stack, and have strengths in operations automation using Python and Bash.

Across 3 companies, I have independently or proactively led cloud migration (AWS → GCP), hybrid architecture design, and on-premises PaaS solution construction, achieving quantitative results including 20-50% cloud cost reduction, 50-67% deployment time reduction, and 90%+ manual work automation.

I aim to continue contributing to infrastructure architecture advancement and organizational productivity improvement.

Strengths

Quick Understanding & Adaptation

- Quickly adapt to new environments and tools, contributing immediately to diverse projects

Systematic Documentation & Knowledge Sharing

- Document all work processes and share with team, contributing to organizational knowledge assets

Record-Based Continuous Improvement

- Manage all tasks including incident response, infra changes, and troubleshooting through records to prevent recurrence and drive improvement

Full-Stack Infrastructure Experience

- End-to-end infrastructure experience spanning cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP
Container Orchestration	Kubernetes (EKS, GKE, On-Premises)
Infrastructure as Code	Terraform, Ansible, Vagrant, Packer
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Jenkins
Monitoring & Logging	PLG Stack, ELK Stack, Thanos
Automation	Go, Python, Bash Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Core Competencies

Cloud Infrastructure Design & Operations

Multi/hybrid cloud architecture design and operations experience on AWS and GCP

- AWS EKS, GCP GKE production operations & cost optimization (monthly \$60,000 → \$48,000, 50% reduction achievements)
- IaC-based infrastructure automation and version control using Terraform (100% GCP resource codification)
- On-premises to cloud hybrid architecture design and migration experience

CI/CD Pipeline & GitOps

Maximizing development productivity through GitOps-based deployment automation

- CI/CD pipeline design and enhancement using GitHub Actions, GitLab CI, ArgoCD
- 50-67% deployment time reduction, 95% rollback time decrease (30min → 1-2min)
- Jenkins-based Unity Android/iOS build automation and Slack-integrated deployment notifications

Monitoring & Observability System Implementation

Establishing proactive incident detection and rapid response systems

- Integrated metric/log monitoring with PLG Stack (Prometheus, Loki, Grafana)
- Centralized logging system with ELK Stack (Elasticsearch, Logstash, Kibana, Filebeat)
- 60-70% incident response time reduction, 99.9% service availability achieved

Automation & Efficiency

Operations automation and developer productivity tools using Python and Bash

- 90%+ manual work reduction through data collection, deployment, and document management automation
- Improved dev team productivity by building internal LLM service (Ollama + Open WebUI)
- Internal tool development: GitLab Webhook doc sync, Slack Bot APK deployment notifications

Professional Experience (Detailed)

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

As a dedicated DevOps infrastructure engineer at a game development company, I design, build, and operate AWS-based cloud and on-premises servers. Development and test servers are on-premises, while QA and Production servers are on AWS, operating a hybrid architecture.

Currently providing secondary technical support and operations for the live game 'Souls' in collaboration with the publisher, and building on-premises/AWS infrastructure to improve development productivity for new game projects.

Project 1: AWS-On-premises Hybrid Cloud Architecture

Contribution 100% (Solo design & implementation) 6 months (2024.10 ~)

Problem

While operating the entire infrastructure on AWS, high cloud costs of \$60,000/month necessitated cost optimization.

Approach

Workload-specific cost analysis revealed that dev/test environments had minimal traffic variation, making cloud elasticity unnecessary. Designed a hybrid architecture transitioning dev/test to on-premises while keeping QA/Production on AWS.

Troubleshooting

- When configuring ALB Ingress on EKS, needed to route to different backends per game client version. Resolved by combining custom header-based routing rules with Target Groups in ALB conditional routing
- Intermittent timeouts during ClusterIP service communication in Kubernetes IPVS mode. Resolved by analyzing Cilium eBPF packet flow and adjusting IPVS connection tracking table settings
- Pod scheduling delays during EFS mounting. Resolved by optimizing EFS CSI Driver mount options and adjusting StorageClass settings

Results

- 20% monthly infra cost reduction (\$60,000 → \$48,000), ~\$144,000 annual savings
- Optimized AWS resource usage by completing on-premises migration of dev/test environments
- Ensured operational stability and prevented failure propagation through environment role separation

Tech Stack

AWS (EKS, EC2, ALB, Route53, ACM, EFS, Aurora MySQL, ElastiCache, CloudFront), Kubernetes, Terraform, Helm

Project 2: On-premises CI/CD Pipeline Construction & Enhancement

Contribution 100% (Full infra & pipeline ownership) 6 months (2024.12 ~)

Problem

Developers manually built and deployed to servers, taking an average of 30 minutes per deployment with frequent failures due to human errors.

Approach

Built a GitOps-based automated deployment environment combining GitLab CI and ArgoCD. Eliminated Docker-in-Docker dependency using Kaniko for Docker image builds, and configured multi-environment (alpha/review/dev/staging) deployments in a single pipeline.

Troubleshooting

- TLS handshake failure with Harbor registry due to OpenSSL version mismatch in Alpine-based containers. Resolved by replacing base image with OpenSSL 3.x compatible version
- Package installation failure on CI Runner due to GPG key expiration after GitLab upgrade. Automated GPG key renewal process to prevent recurrence
- Build time spikes due to cache invalidation during Kaniko builds. Resolved with `-cache=true --cache-ttl=24h --snapshot-mode=redo` option combination

Results

- 67% deployment time reduction (30min → 10min), 90% manual work automation
- 3x+ weekly deployments (1-2 → 5-7)

Tech Stack

GitLab CI/CD, ArgoCD, Jenkins, Kaniko, Kubernetes, Docker, Harbor

Project 3: ELK-based Integrated Monitoring & Logging System

Contribution 100% (System design & implementation) 4 months (2025.06 ~)

Problem

Server and container logs were distributed across the on-premises environment, making root cause analysis time-consuming during incidents.

Troubleshooting

- Logstash parsing failures due to null bytes in game server logs. Resolved by pre-removing with mutate filter `gsub`
- Elasticsearch field mapping conflicts. Resolved with explicit index template mappings and type conversion filters
- Index read-only due to disk watermark exceeded. Stabilized with ILM policy auto-deleting indices older than 7 days

Results

- 90% log search time reduction, 70% root cause analysis time reduction
- Real-time processing of 20GB daily log data with 7-day retention

Tech Stack

Elasticsearch, Logstash, Kibana, Filebeat, Kubernetes

Project 4: Developer Productivity Tools (Doc Automation · LLM Service · APK Deploy Bot · Git Mirroring)

Contribution 100% 2024.12 ~ 2026.01

4-1. GitLab-Google Drive Document Automation (2 weeks, 2025.08)

Auto-sync on push via GitLab Webhook + Google Drive API. 80% document management time reduction

4-2. Internal LLM Service (4 weeks, 2026.01)

Deployed Ollama + Open WebUI on Kubernetes on on-premises GPU server (RTX 3060). Reduced external LLM costs and eliminated data leakage risks

4-3. Slack APK Distribution Bot (4 weeks, 2025.02)

Auto-notification to Slack with QR code on Jenkins build completion. 90% QA team delivery time reduction

4-4. Git Repository Mirroring Tool Development (4 weeks, 2026.02)

Developed Go-based bidirectional Git mirroring tool (git-bridge). Auto-sync across CodeCommit/GitLab/GitHub, 100% multi-platform mirroring automation

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Worked as a DevOps Engineer at a game and blockchain startup, leading the large-scale cloud migration project from AWS to GCP.

Project 1: Large-scale AWS → GCP Cloud Migration & CI/CD Reconstruction

Contribution 70-100% 7 months (2023.05 ~ 2023.12)

Troubleshooting

- Routing issues during AWS ALB → GCP Global LB transition. Reconfigured after analyzing GCP NEG structure
- Data corruption during RDS → Cloud SQL migration due to character set differences. Resolved with pre-validation scripts and step-by-step integrity checks
- Network design issues due to AWS/GCP subnet model differences. Redesignated CIDR and converted to GCP tag-based firewalls

Results

- 50% cloud cost reduction (\$10,000 → \$5,000/month), 50% deployment time reduction, 95% rollback reduction
- Full GCP infrastructure Terraform IaC, 100% deployment history tracking via Git commits

Tech Stack

GCP (GKE, Cloud SQL, Cloud Storage, VPC, Global LB), Terraform, GitLab CI, ArgoCD, GitHub Actions, Helm

Project 2: Game Data Collection Automation System

Contribution 100% 3 months (2024.01 ~ 2024.03)

Results

- 95% data collection automation, 0% data gap rate, 10x throughput increase

Tech Stack

Python, Cloud Functions, Cloud Scheduler, Google Sheets API

Project 3: PLG Stack Monitoring System

Contribution 100% 1 month (2024.04)

Results

- 80% proactive detection, 60% response time reduction, availability 99.5% → 99.9%

Tech Stack

Prometheus, Loki, Grafana, Promtail, Fluent-bit, Slack API

Iorchard

2022.04 ~ 2023.03

Infra Engineer

Built PaaS (Kubernetes + OpenStack + Ceph) infrastructure solutions for clients. Designed on-premises cloud infrastructure for financial and public sector clients, conducted operations training.

Key Projects: PaaS Solution · DP Training · Certificate Automation

Results

- PaaS solution for 5 clients, Kubernetes HA 99.9% availability
- Established DP's autonomous Kubernetes operations (6-month training)
- 10x certificate renewal extension (1yr → 10yr), 50hrs annual ops savings

Tech Stack

Kubernetes, OpenStack, Ceph, Ansible, kubeadm, Rocky Linux/CentOS

ETech System

2022.01 ~ 2022.04

System Engineer

Hardware Server and SAN storage construction. Transitioned to DevOps field for career growth.

Opensource Projects

Kubernetes CRD

[k8s-namespace-sync](#) — Automatically sync Kubernetes namespace resources across clusters

[helios-lb](#) — Custom load balancer controller for Kubernetes

[network-policy-generator](#) — Kubernetes NetworkPolicy auto-generation controller

GitHub Actions

[Compress Decompress Action](#) — Compress/decompress files in workflows

[Image Tag Updater](#) — Auto-update image tags in YAML/Helm files

[Go Git Commit Action](#) — Go-based Git commit creation and push

[Extract Commit Action](#) — Extract Git commit SHA, message, and author information

[Multi Git Mirror Action](#) — Automate mirroring across multiple Git repositories

[Kube Diff Action](#) — Compare and report Kubernetes resource changes

Ansible Galaxy

[Ansible K8s IAC Tool](#) — Automated K8s and IaC tool installation collection

[Ansible User Management](#) — Linux user and SSH key management role

DevOps Tools

[bash-pilot](#) — AI-powered Bash command assistant CLI tool

[git-bridge](#) — Git repository mirroring and sync CLI tool

Chrome Extensions

[Dev Toolkit](#) — Developer utility collection

[QRify](#) — Instant QR code generation for URLs/text

Education

Chungnam National University - Public Administration (Transfer)

2018.03 ~ 2020.08

National Institute for Lifelong Education - Business Administration (Academic Credit Bank)

2016.09 ~ 2018.02

Certifications

Engineer Information Processing

2021.11

AWS Certified Solution Architect - Associate (Expired)

2021.11

Network Administrator Level 2

2021.10

Linux Master Level 2

2021.10

Computer Specialist Level 1

2021.04

TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)

2021.02

Study

[T101\(Terraform\) Study](#)

2023.08 ~ 2023.10

[AEWS\(EKS\) Study](#)

2023.04 ~ 2023.06

[PKOS\(Kubernetes\) Study](#)

2023.01 ~ 2023.02

[Cloud Security Training](#)

2022.02 ~ 2022.04

Security Solution Operation Specialist Training Program

2021.07 ~ 2021.12